Mental Health Monitoring Grand Challenge

Accelerating the Development of Multimodal Mental Health Tracking in Japan

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Executive Summary

Japan faces rising depression, social withdrawal (*hikikomori*, 引きこもり), death from overwork (*karoshi*, 過労死), and suicide, along with a low-birthrate and aging-population crisis (*shoshi koreika*, 少子高齢化). Mental health causes large human and economic loss, but it is still tracked through questionnaires and occasional clinical interviews. These tools are useful, but they are imprecise, expensive, and slow. They do not give enough real-time information for early prevention, individualized treatment planning, or population-scale monitoring. Many research groups in Japan have already begun to explore how cheap and widely available technologies such as phones and wearables can be used for detecting mental states in real-time, but progress is disjoint and slow. Each group uses different participants, measures different signals, and targets different mental health problems. How do we bring together all these approaches to create a solution that accurately monitors multiple important mental health outcomes using consumer technologies? Inspired by the DARPA Self-Driving Car Grand Challenge, we propose launching a Mental Health Monitoring Grand Challenge where up to 20 research groups in Japan compete to solve this question for a prize. Over three rounds of competition, research groups would be asked to accurately identify mental health conditions using multimodal data from phones and consumer wearables. Each round will become more difficult to reward solutions that account for privacy and user satisfaction. About 1.5B JPY would be needed to support research teams, recruit human participants, and build data collection infrastructure. If successful, this program would accelerate research to solve one of Japan's most pressing problems.

Heilmeier Question	**Answer**
What are you trying to do?	Develop and validate practical methods for continuously measuring mental health in everyday life using smartphones, wearables, behavioral data, and other easy-to-collect signals.
How is it done today?	Mental health is measured mainly through questionnaires and clinical interviews. These methods provide limited information between assessments and are difficult to scale. There is new research to detect mental health conditions with apps, wearables, and AI but it is still limited.
What is new?	A research prize challenge for multimodal detection of multiple mental health conditions via consumer technologies. Data and participants are standardized to allow for many research groups to compete under difficult, real-world conditions.
Who cares?	Better monitoring could help clinicians customize patient treatment, researchers evaluate interventions, employers assess workplace programs, governments track population trends, and individuals monitor their own mental health.
What are the risks?	Recruitment of human participants with varied mental health conditions poses the greatest uncertainty.
How much will it cost?	JPY 1.5B in funding for the inaugural Japan challenge.
How long will it take?	Three years. One year to set up challenge infrastructure, followed by two years of the challenge.
What are the milestones?	Recruit participants, collect the shared-data benchmark, run live trials with finalists, and release carefully anonymized public research data.

[*] This is a condensed concept note. A full technical dossier is available for potential funders upon request. It includes the detailed budget model and line items, literature reviews of mental-health measurement approaches, the Japanese research ecosystem map, metric design arguments, and the staged validation model behind this program.

1. Why Progress Is Stuck

Mental-health conditions are common, costly, and hard to measure. In Japan, clinicians and researchers often use self-report screening surveys, such as the PHQ-9 for depression or the GAD-7 for anxiety, which ask people to report their recent symptoms. These surveys are useful, but they are coarse, subjective, and burdensome to repeat. They are mainly used for screening and symptom tracking, not formal diagnosis. Formal diagnosis relies on clinical interviews conducted by trained professionals using international standards such as ICD and DSM. These interviews are more accurate, but they are time-consuming, infrequent, and limited by the number of trained clinicians. More recent work has emerged around fMRI for diagnosis, but that too is expensive and hard to scale. The field needs better ways to measure mental health accurately, continuously, affordably, and seamlessly.

The Big-if-True Bet

Measuring *multiple* mental health conditions can be done *simultaneously* by combining multimodal signals from consumer technologies such as phones, social media, and wearables using AI.

In Japan, several groups are already developing objective and scalable approaches to mental-health measurement. Keio University researchers use wristband wearable data, including activity, sleep, heart rate, and skin temperature, to screen for depression and estimate symptom severity. Sophia University and Osaka Metropolitan University researchers have estimated anxiety and stress using smartphone, social-media, and Oura ring data. Institute of Science Tokyo researchers use smartphone usage logs to study problematic smartphone use. At the National Center of Neurology and Psychiatry, researchers use tablet-based cognitive tasks and eye-movement measures to assess schizophrenia.

Japan already has strong components of a mental-health measurement ecosystem, but these efforts remain fragmented. Each effort uses different signals, participants, and outcomes, making the results difficult to compare or combine. This fragmentation is a problem because mental health is interconnected. Depression, anxiety, addiction, and other conditions often overlap. To capture a fuller picture of mental health, these separate pieces need to be brought together.

Key Bottleneck

Scientific progress is limited by fragmentation, coordination costs, and barriers to validation. Researchers often study small slices of mental health because larger studies are risky, expensive, and hard to run. A Grand Challenge can turn many partial solutions into a cohesive solution. Funding and prize money make the harder problem worth pursuing. Shared challenge infrastructure lowers the practical burden of participation and makes results easier to compare.

This situation resembles the state of research for autonomous driving before the DARPA self-driving vehicle challenges. Many groups had solved important pieces, including sensing, mapping, planning, and control, but the field lacked a way to combine them into systems that worked in the real world. DARPA helped by defining a concrete task, creating shared testing conditions, and rewarding teams that could put the pieces together.

Mental Health Monitoring Challenge

We propose taking a DARPA Grand Challenge approach to mental health monitoring. Given one week of data from smartphones and commercially available wearables, teams compete to predict which mental-health conditions a participant is experiencing and how severe those symptoms are. The best systems would be accurate, low-burden, and privacy-preserving. The challenge would handle participant recruitment, data collection, clinical labeling, and evaluation, so teams can focus on the machine learning problem. Our goal is to move the field from promising pieces toward integrated systems that can measure mental health in the wild.

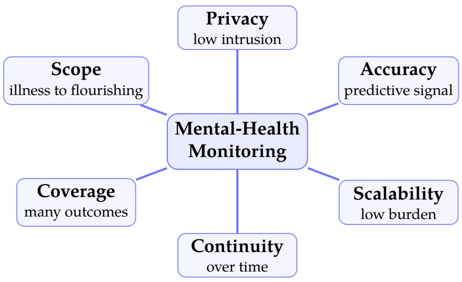
Better measurement changes what the field can see and act on. In diabetes care, glucose monitoring helps identify when a person needs timely intervention, such as medication, food, or behavioral adjustment. In weather forecasting, continuous measurement at scale makes it possible to detect patterns, predict future conditions, and prepare for risks such as natural disasters. Mental health needs a similar shift. At the individual level, easier, more accurate, and more continuous measurement would help clinicians track how symptoms change, detect risks earlier, and evaluate whether treatment is working. At the population level, it would help researchers and policymakers identify trends, compare interventions, and direct resources where support is needed.

2. A Common Framework for Mental-Health Monitoring

Mental-health monitoring technologies differ enormously. Some rely on symptom questionnaires, others use smartphones, wearable sensors, cognitive tests, speech, eye movements, or physiological signals. Each captures a different aspect of mental health and reports performance using different metrics. As a result, it is difficult to compare approaches directly or determine which systems represent genuine progress.

Rather than evaluating technologies by the sensors they use, we propose evaluating them by the capabilities they provide. An ideal mental-health measurement system should accurately measure a person's mental state while remaining practical enough for routine use in everyday life.

This perspective suggests six desirable properties:



Metric	Selects for...
Privacy	Can it work without collecting data that feels intrusive or unsafe?
Accuracy	Does it predict real mental-health states and outcomes?
Scalability	Is it cheap, low burden, and practical for many people?
Continuity	Can it monitor people over time, not only during clinic visits?
Coverage	Can it track several conditions and related outcomes?
Scope	Can it measure illness, recovery, resilience, and flourishing?

Figure 1: Six dimensions for evaluating mental-health measurement systems.

These properties define a common design space for mental-health measurement. Existing approaches occupy different parts of this space. For example, clinical interviews are the current gold-standard for diagnosis but are expensive and difficult to scale. Questionnaires are inexpensive but rely on self-report and provide only occasional snapshots. Meanwhile, smartphones and wearable devices enable continuous measurement with relatively little effort, but their accuracy and clinical validation still need to improve. No existing approach simultaneously optimizes all six dimensions.

What Success Looks Like

The goal of mental health monitoring is not solely to predict clinical outcomes accurately. The best mental-health monitoring systems should combine strong accuracy with privacy, scalability, continuous monitoring, broad clinical coverage, and, ultimately, the ability to measure the full spectrum of mental well-being from individual to population scale.

3. Inaugural Challenge

The inaugural challenge asks teams to predict weekly severity across several mental-health domains from smartphone, wearable, and easy-to-collect behavioral data. The launch domains are depression, anxiety, sleep disturbance, burnout or stress, and social withdrawal. This design reflects how mental health appears in real life where symptoms overlap, causes are mixed, and useful measurement systems need to read several outcomes from the same everyday data.

Inaugural Grand Challenge

Build systems that predict weekly severity across at least five mental-health domains from wearable, smartphone, and easy-to-collect behavioral data. Phase 1 uses a shared standard dataset. Phase 2 adds richer data and privacy-weighted scoring. Phase 3 tests five finalist teams with live participants.

The challenge becomes harder in three phases. Phase 1 establishes a broad benchmark using a shared standard dataset. Phase 2 adds privacy-weighted scoring, so teams are rewarded for accurate predictions that rely on data participants are more willing to share. Phase 3 tests finalists with live participants and team-designed apps, where burden, missing data, engagement, and real-world use become part of performance.

The launch challenge will include 2,000 participants, divided across three separate phases. The budget assumes broad entry in Phase 1, 10 funded teams in Phase 2, and five finalists in Phase 3.

Phase	People	Teams	Core test	Selects for…
1	500	Broad entry	Shared dataset with held-out scoring	Strong passive prediction.
2	500	10 (funded)	Richer data with privacy-weighted scoring	Accuracy with acceptable data.
3	1,000	5 finalists	Live app test with separate cohorts	Real-world performance.

The competition also includes a scientific discovery track. A subset of participants would receive deeper measurement, such as structured interviews, cognitive tasks, physiological assays, biomarkers, neuroimaging, or longer follow-up. These measures can stay outside the competitor data package. Their purpose is to help scientists interpret the lower-cost signals and build a research resource for the field.

Milestones

The midterm milestone is a completed Phase 1 benchmark, including collected data, clinical labels, hidden scoring, team submissions, and a safely anonymized public dataset. Final success means that teams accurately predict several mental-health domains in new participants, beat simple questionnaire and demographic baselines, respect participant privacy preferences, maintain real-world use, along with the release of a broader public research dataset.

4. Budget and Timeline

The inaugural challenge requires JPY 1.5B in funding. The competition produces final results in 16–20 months. The scientific track continues through Month 28 because deeper measurement, analysis, and public resource preparation continue after the winners are selected.

Budget item	Funding	Timing
Core operations	JPY 682M	Months 0–20
Team incentives and prizes	JPY 600M	Months 6–20
Scientific research track	JPY 75M	Months 6–28
Contingency	JPY 143M	Full program
Total Budget	**JPY 1.5B**	**Full program**

Program step	Commitment	Timing
Setup	–	Months 0–6
Phase 1	500 people	Months 7–9
Phase 2	500 people	Months 10–13
Finalist build period	5 finalists	Months 13–16
Phase 3	1,000 people	Months 16–18
Final results	–	Months 19–20
Scientific closeout	100-person cohort	Months 21–28

5. Why Japan

Japan is an ideal place to launch this challenge because the national need, scientific opportunity, and research ecosystem are uniquely aligned. Rising depression, social withdrawal (*hikikomori*, 引きこもり), death from overwork (*karoshi*, 過労死), and suicide compound Japan's low-birthrate and aging-population crisis (*shoshi koreika*, 少子高齢化), making the loss of human potential increasingly costly. Better mental health measurement would enable earlier detection of withdrawal, exhaustion, relapse risk, and recovery, allowing institutions to intervene sooner and evaluate which interventions work.

Japan is also well positioned to lead scientifically. Many psychiatric assessment tools were developed primarily in Western populations, so a Japanese grand challenge would help establish measurement systems that generalize across cultures, languages, and social contexts. At the same time, Japan already has strong research groups developing wearable sensing, smartphone monitoring, brain biomarkers, workplace digital mental health, and behavioral assessment. The missing ingredient is a common framework for evaluation. The Mental Health Monitoring Grand Challenge would provide shared datasets, validation standards, and incentives to integrate these efforts into the first scalable infrastructure for objective mental health measurement.